

BATCH DATA ENGINEERING PROJECT REPORT

ServiceTrack

Job Tracking & Customer Visit Analytics Pipeline

Python • PySpark • Databricks • Delta Lake • SQL • Power BI



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Abstract

The ServiceTrack: Job Tracking & Customer Visit Analytics Pipeline is a batch data engineering project developed using the Medallion Architecture, comprising Bronze, Silver, and Gold layers. The project processes raw service center data and transforms it into clean, structured, and analytics-ready datasets suitable for enterprise reporting.

The pipeline was implemented using Python, PySpark, Databricks, Delta Lake, SQL, and Power BI. During the data processing stage, duplicate records were removed, missing values were handled, and additional business metrics such as repair duration and delivery status were generated to enrich the dataset.

The final Gold layer contains multiple business-ready tables that answer important operational questions related to technician performance, repair delays, device brand trends, repeat customers, repair cost analysis, issue analysis, and monthly service trends. These tables were connected directly to Power BI to build an interactive dashboard for business users.

This project demonstrates how raw operational data can be converted into meaningful business insights using a modern, production-style data engineering pipeline.

Project Objectives

- Build a complete batch data engineering pipeline from raw data to business insight.
- Implement the Medallion Architecture (Bronze, Silver, Gold) on Databricks.
- Clean, standardize, and transform raw service center data.
- Perform business-oriented analysis across technicians, devices, and customers.
- Store all processed data reliably in Delta Lake tables.
- Generate Gold layer tables covering the full set of key business metrics.
- Connect Databricks Gold tables directly with Power BI.
- Build an interactive, decision-ready dashboard for business users.
- Demonstrate a real-world, end-to-end data engineering workflow.

Technology Stack

The pipeline was built entirely on the Databricks Lakehouse platform, combining distributed processing, reliable storage, and enterprise-grade visualization in a single connected workflow.

Technology	Purpose
Python	Core scripting and data processing logic
PySpark	Distributed data transformation across all layers
Databricks	Cloud development and orchestration environment
Delta Lake	Reliable, versioned storage for Bronze/Silver/Gold tables
SQL	Business queries and Gold layer aggregations
Power BI	Dashboarding and interactive visualization
GitHub	Source control and project versioning

Medallion Architecture

The pipeline follows the industry-standard Medallion Architecture, progressively refining data through three layers before it reaches business users via Power BI.



BRONZE LAYER

- Loaded raw CSV files into Databricks.
- No transformations were applied at this stage.
- Original source data was fully preserved for traceability.

SILVER LAYER

- Removed duplicate records.
- Filled missing technician names.
- Converted and standardized date columns.
- Calculated repair duration for every job.
- Created a derived delivery status field.
- Joined customer, device, and service tables into a unified dataset.

GOLD LAYER

Created business-ready analytical tables covering technician performance, delay analysis, brand analysis, monthly repairs, warranty analysis, job status, issue analysis, repair cost, business summary, repeat customers, technician workload, and city analysis.



Data Cleaning Report

Activity	Status
Raw CSV Imported	Completed
Duplicate Records Removed	Completed
Missing Technician Names Filled	Completed
Date Format Corrected	Completed
Repair Duration Created	Completed
Delivery Status Created	Completed
Silver Table Created	Completed
Gold Tables Generated	Completed

DATA QUALITY SUMMARY

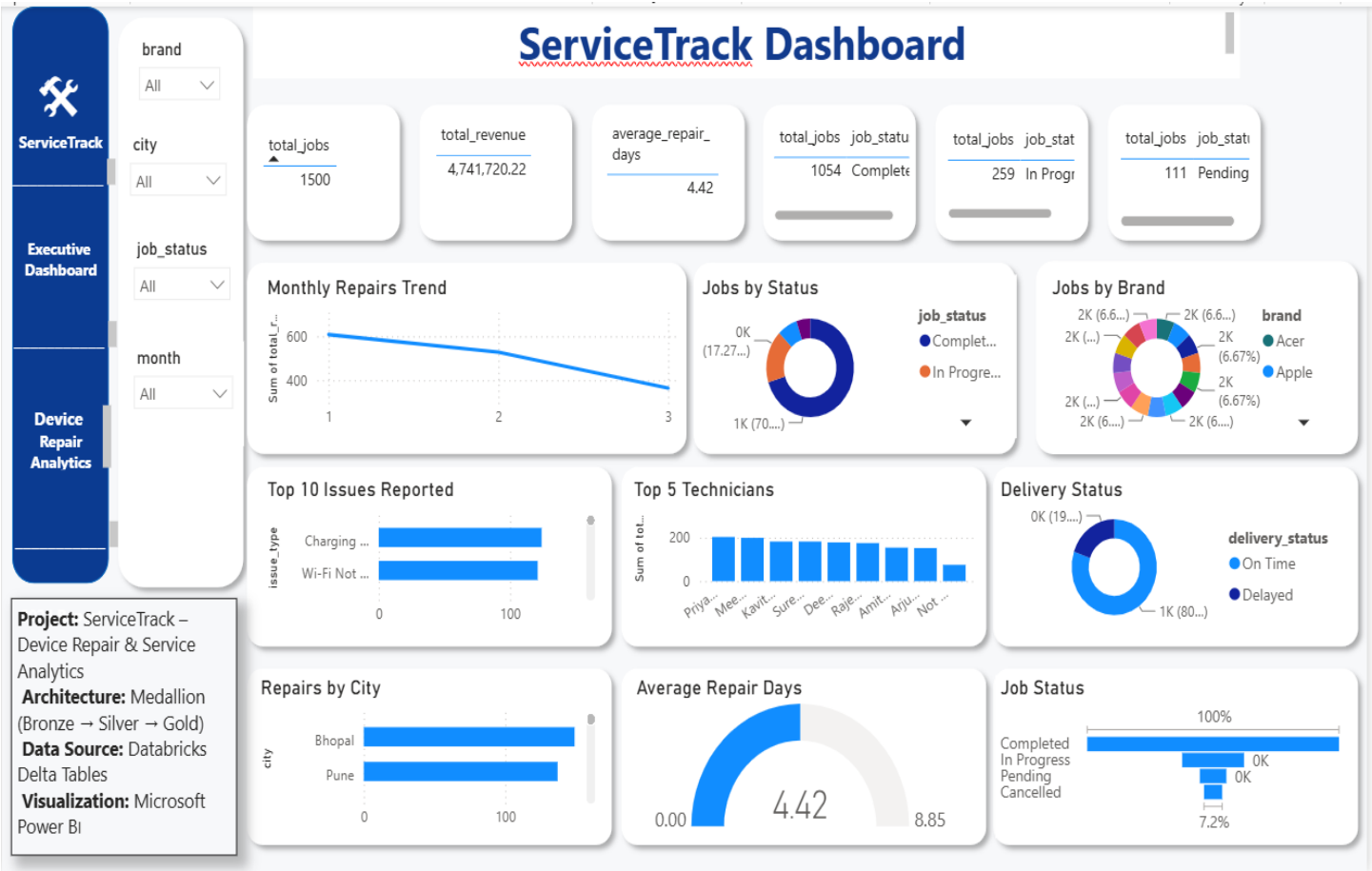
The raw dataset contained duplicate records and missing technician names. During the Silver Layer stage, duplicate rows were removed, missing values were handled, and date columns were converted into proper formats. New business columns such as Repair Duration and Delivery Status were generated to support further analysis. These transformations ensured that the Gold Layer contained clean, reliable, and analytics-ready data.

Gold Layer Tables

Twelve Gold layer tables were generated, each answering a distinct operational or business question. Together they form the semantic layer that powers the Power BI dashboard.

Gold Table	Purpose
gold_business_summary	Overall KPIs
gold_technician_performance	Technician efficiency
gold_delay_analysis	Delay statistics
gold_brand_analysis	Brand trends
gold_issue_analysis	Common issues
gold_monthly_repairs	Monthly trends
gold_repair_cost	Cost analysis
gold_repeat_customers	Repeat customers
gold_city_analysis	City analysis
gold_warranty_analysis	Warranty jobs
gold_job_status	Job status
gold_technician_workload	Technician workload

Dashboard Overview



The dashboard was connected directly to the Databricks Gold Layer tables and provides interactive visualizations for business users. It includes KPI cards, slicers, bar charts, donut charts, gauge charts, and trend analysis to monitor service center performance end-to-end.

Key Business Insights

- Successfully processed 1,500 service records end-to-end.
- Average repair duration across all jobs was 4.42 days.
- Generated 12 business-ready Gold layer tables for reporting.
- Identified technician workload distribution across the service center.
- Analyzed device brand repair trends to surface reliability patterns.
- Identified repeat customers for retention-focused insights.
- Monitored repair costs and monthly repair trends over time.
- Built an interactive Power BI dashboard to support decision making.

Challenges Faced

During the implementation of this project, several practical challenges were encountered. Understanding the Medallion Architecture and the Databricks environment was initially difficult, as this was my first hands-on data engineering project. Managing multiple Gold layer tables, handling missing values, creating Delta tables, and connecting Databricks with Power BI required careful, deliberate implementation.

Debugging Spark transformation errors and validating the correctness of transformed data at each layer were also important and valuable learning experiences that shaped my understanding of production-grade data pipelines.

Learning Outcomes

Through this project, I gained practical experience in building a complete batch data engineering pipeline using Databricks and PySpark. I learned how to implement the Medallion Architecture, perform structured data cleaning, create and manage Delta tables, and write both SQL and PySpark transformations to derive business-ready datasets.

I also learned how to integrate Databricks with Power BI and design an interactive dashboard that translates raw operational data into actionable business analysis for non-technical stakeholders.

Conclusion

The ServiceTrack pipeline successfully transformed raw operational data into structured analytical datasets using a modern Medallion Architecture. The project demonstrates the complete lifecycle of data engineering, from ingestion and transformation to visualization.

The Power BI dashboard enables business users to monitor service center performance through interactive reports and meaningful insights. Overall, this project provided valuable, practical experience in modern data engineering technologies and analytics, and reflects a production-style approach to solving real business problems with data.